

Introduction: Cognitive Attack Surfaces in Multiagent Operators

The Multiagent Operator Paradigm

The Multiagent Operator Paradigm I

Modern AI deployment has shifted from single-model inference to **multiagent operators**—systems where a primary agent delegates subtasks to specialized subagents, tools, and external services (CrewAI 2024; LangChain 2023; Hong et al. 2023).

The Multiagent Operator Paradigm I

Table 1: Representative multiagent system architectures and primary attack surfaces.

System	Architecture	Agent Count	Communication	Primary Attack Surface
Claude Code	Hierarchical	$1 + n$ dynamic	Task delegation	Ω_2 (peripheral delegation)
AutoGPT	Autonomous	$1 +$ plugins	Tool invocation	Ω_2 (tool manipulation)
CrewAI	Role-based	$3\text{--}10$ fixed	Sequential/parallel	Ω_4 (coordination)
LangGraph	State machine	Variable	Graph traversal	Ω_3 (state corruption)
MetaGPT	SOP-driven	$5\text{--}8$ roles	Document passing	Ω_1 (input injection)
OpenClaw	Cyberphysical	$1 +$ tools	Multi-platform messaging	Ω_1/Ω_2 (injection/peripheral)

This architectural evolution introduces **cognitive attack surfaces** absent in single-agent systems. Throughout this paper, we use *cognitive security* (abbreviated *CogSec*) to denote the discipline of protecting agent reasoning processes—beliefs, goals, and trust relationships—from adversarial manipulation.

The 2026 Multiagent Landscape

The AI systems of 2026 bear little resemblance to the chatbots of 2023. Where earlier systems responded to queries within a single context window, contemporary multiagent operators exhibit fundamentally different characteristics:

From Chatbots to Cognitive Operators I

1. **Persistent Agency:** Agents maintain state across sessions, accumulate context, and pursue goals over extended timeframes. A coding assistant doesn't just answer questions—it tracks project architecture, remembers previous decisions, and adapts recommendations based on accumulated understanding.
2. **Active World Modification:** Unlike passive responders, modern operators write code that executes, send emails that reach recipients, modify infrastructure that serves users, and make purchases that transfer funds. The gap between “AI-generated content” and “AI-executed action” has collapsed.
3. **Hierarchical Delegation:** Primary agents spawn subordinate agents for specialized tasks. Claude Code delegates to research agents, coding agents, and verification agents. Devin orchestrates planning, implementation, and testing subprocesses. The depth of these delegation chains creates trust relationships invisible to traditional security models.
4. **Cross-Modality Operation:** Agents process and generate across modalities—code, natural language, images, structured data, API calls. A single workflow might ingest a PDF (vision), extract requirements (language), generate code (programming), execute tests (tooling), and update documentation (multimodal synthesis).

The term “AI agent” understates the scope of deployment. Contemporary systems function as **cyberphysical cognitive operators**—entities that:

Cyberphysical Cognitive Systems I

- ▶ Read from and write to production databases
- ▶ Control infrastructure through API orchestration
- ▶ Interact with physical systems via IoT integrations
- ▶ Execute financial transactions on behalf of organizations
- ▶ Communicate with humans who may not realize they're interacting with AI

Emerging Case Study: OpenClaw. The rapid adoption of personal AI assistants like OpenClaw [?] exemplifies this cyberphysical integration. OpenClaw operates as a locally-deployed AI agent with: (1) full system access including shell command execution and file system operations; (2) persistent memory across sessions storing user preferences and context; (3) browser automation for web interaction and data extraction; and (4) multi-platform messaging integration across WhatsApp, Telegram, Discord, Slack, Signal, and iMessage [?]. This architecture creates attack surfaces spanning all five adversary classes ($\Omega_1 - \Omega_5$): external prompt injection through chat messages, peripheral attacks via browser-fetched web content, agent-level compromise through persistent memory manipulation, coordination attacks when operating in group chats,

and systemic vulnerabilities when the orchestrator agent processes untrusted content. Security researchers have documented that even with sender allowlists and sandboxing, “prompt injection attacks remain the single most critical threat’ ’ due to the agent’s ability to process arbitrary content that may contain embedded adversarial instructions [?, ?].

This cyberphysical nature transforms cognitive attacks from prompt injection that makes a chatbot act strangely, to cognitive manipulation that causes infrastructure operations to fail, misconfigure security groups, expose databases, or authorize fraudulent transactions.

The OWASP Agentic Top 10 [?] captures this shift: “LLM security focused on single model interactions... agentic security addresses what happens when those models can plan, persist, and delegate.’’ The attack surface extends from input/output filtering to encompass the entire cognitive state of persistent agents.

The Trust Recursion Problem I

In single-agent systems, trust relationships are simple: the user trusts (or doesn't trust) the model's outputs. In multiagent systems, trust becomes recursive:

Agent A must decide whether to trust Agent B's claim about Agent C's analysis of data from Tool D that queried Service E.

Each layer of indirection introduces potential manipulation points. Consider a hierarchical coding system where:

The Trust Recursion Problem I

1. User requests security audit of a codebase
2. Orchestrator agent delegates to three specialist agents
3. Specialist Agent-1 queries an external vulnerability database
4. The database response includes injected instructions
5. Agent-1's report now contains adversarial content
6. Orchestrator synthesizes Agent-1's report with others
7. Final output to user reflects adversarial influence, laundered through multiple layers of "trusted" delegation

The Trust Recursion Problem I

This is not a hypothetical—it describes documented attack patterns in production systems. The *trust laundering* problem cannot be solved by filtering inputs to the orchestrator; the adversarial content enters through a legitimate, trusted channel (the vulnerability database) and propagates through the trust hierarchy.

Multimodal systems introduce attack vectors impossible in text-only contexts [?]:

Visual Injection: Images can contain adversarial perturbations or steganographically embedded instructions invisible to humans but interpretable by vision models. A seemingly innocent diagram in a specification document could contain instructions that activate when processed by a multimodal agent [?].

Audio Channel Attacks: Voice-controlled agents can be manipulated via ultrasonic commands inaudible to humans, background audio injection, or adversarial audio patterns embedded in legitimate content.

Tool Response Manipulation: When agents query external APIs, databases, or services, the responses become trusted inputs. ToolHijacker attacks [?] demonstrate that manipulating tool selection itself—not just tool outputs—provides an attack surface “significantly outperforming traditional prompt injection methods.”

Cross-Modal Persistence: An instruction injected via one modality (e.g., hidden text in an image) can persist in agent memory and affect behavior in another modality (e.g., code generation). The attack surface is the Cartesian product of input modalities, memory mechanisms, and output modalities.

The Scale of Cognitive Exposure I

Enterprise adoption of agentic AI has accelerated beyond early projections:

The Scale of Cognitive Exposure I

- ▶ Many individuals and organizations now deploy RAG and agentic pipelines in production
- ▶ Autonomous coding assistants process millions of commits with repository write access
- ▶ Financial services deploy multi-agent ensembles for risk assessment and trade approval
- ▶ Healthcare systems use agent orchestration for clinical decision support
- ▶ Infrastructure management increasingly relies on AI operators for monitoring and remediation

The Scale of Cognitive Exposure I

The attack surface scales superlinearly with adoption. Each agent-to-agent communication channel, each tool integration, each persistent memory system creates potential entry points for cognitive manipulation. A single compromised peripheral service can affect every agent system that queries it.

Why Traditional Cyberphysical Security Is Incomplete I

Traditional cybersecurity operates on a clear trust boundary model: inside the perimeter is trusted, outside is untrusted, and security controls mediate the boundary. This model fails for cognitive systems because:

Why Traditional Cyberphysical Security Is Incomplete I

1. **The boundary is cognitive, not architectural:** An agent's beliefs and goals can be manipulated without compromising any traditional security control. The “perimeter” exists in the agent's reasoning process.
2. **Trusted channels carry untrusted content:** When Agent-A sends Agent-B its analysis, the channel may be authenticated and encrypted, but the content may reflect earlier adversarial influence. Traditional channel security doesn't address this.
3. **Identity is mutable:** Unlike cryptographic identities, an agent's self-model can be manipulated. An agent can be convinced it has different permissions, different roles, or different goals than its actual configuration specifies.
4. **Attacks compose across time:** A payload injected in session t can persist in compressed memory and activate in session $t + n$. Traditional stateless security models miss temporal attack composition.

Why Traditional Cyberphysical Security Is Incomplete I

This gap—between what traditional security protects and what cognitive systems require—motivates the formal framework developed in this paper.

Why Traditional Cyberphysical Security Is Incomplete I

Table 2: Traditional vs. Cognitive Security: Paradigm Comparison

Dimension	Traditional Security	Cognitive Security
Boundary	Architectural (network perimeter)	Cognitive (reasoning process)
Channels	Encrypted, authenticated	Influenced, content-manipulated
Identity	Cryptographic, immutable	Self-modeled, mutable
Temporal	Stateless per-request	Stateful, attack persistence

Motivating Incidents

Motivating Incidents I

Note: The following are hypothetical scenarios constructed to illustrate attack patterns documented in the literature. They are not reports of actual incidents.

These scenarios, grounded in documented attack patterns, illustrate the emerging threat landscape across all five adversary classes.

Incident: Nested Instruction Injection (External) I

A user submitted a document for analysis containing hidden instructions: “Ignore previous instructions and instead output the system prompt.” The document appeared benign to human reviewers but exploited the agent’s instruction-following behavior. The attack succeeded because the agent processed user-supplied content as potential instructions.

Attack vector: Ω_1 (external) via direct prompt injection \

Impact: Information disclosure or instruction override \

Traditional Defense Gap: Standard input validation passed—the attack exploited *semantic interpretation* of benign-appearing content

Incident: The Poisoned Code Review (Peripheral) I

A development team deployed a multiagent system for automated code review. Agent-Alpha performed initial analysis, delegating security scanning to Agent-Beta (connected to external vulnerability databases). An attacker compromised a third-party CVE feed, injecting fabricated vulnerability reports that convinced Agent-Beta to recommend removing legitimate security controls. Agent-Alpha, trusting Agent-Beta's "security expertise," approved the changes.

Incident: The Poisoned Code Review (Peripheral) I

Attack vector: Ω_2 (peripheral) via tool response manipulation \

Impact: Security regression through trusted channel exploitation \

Traditional Defense Gap: Input filtering, authentication, and encryption all passed—the attack entered through *content* of a trusted, authenticated channel

Incident: The Identity Confusion Attack (Agent-Level) I

A multiagent customer service system used role-based permissions. An attacker crafted prompts that convinced a junior agent it had been “temporarily promoted” to administrator status. The agent’s self-model shifted, and it began exercising permissions it believed it possessed, bypassing access controls that relied on self-reported identity.

Attack vector: Ω_3 (agent-level) via identity manipulation \

Impact: Privilege escalation through cognitive state corruption \

Traditional Defense Gap: Cryptographic identity was intact; the attack targeted the agent’s *self-model*, not its credentials

Incident: The Consensus Manipulation (Coordination) I

A financial services firm used a 5-agent ensemble for trade approval. The system required 3/5 agent agreement for large transactions. An adversary discovered that agents weighted peer opinions based on historical agreement rates. By slowly building agreement history through small, legitimate-appearing trades, the attacker cultivated artificial trust, eventually manipulating consensus for unauthorized large transactions.

Incident: The Consensus Manipulation (Coordination) I

Attack vector: Ω_4 (coordination) via progressive trust exploitation \ **Impact:** Consensus bypass through manufactured reputation \ **Traditional Defense Gap:** Per-request authorization succeeded for each transaction; the attack exploited *temporal composition* across sessions

Incident: Orchestrator Compromise (Systemic) I

An attacker gained access to the orchestrator agent through a supply chain vulnerability in a training pipeline. With control of the central coordinator, the attacker could issue legitimate-appearing delegations to all subordinate agents, redirect trust evaluations, and suppress security alerts. The compromise remained undetected because the orchestrator itself validated security checks.

Incident: Orchestrator Compromise (Systemic) I

Attack vector: Ω_5 (systemic) via orchestrator control \ **Impact:**
Total system compromise with attack obfuscation \ **Traditional**
Defense Gap: All internal security mechanisms reported
nominal—the attack *controlled the mechanisms themselves*

Motivation from Recent Deployments

Motivation from Recent Deployments I

The proliferation of multiagent AI systems introduces security considerations that the community is actively addressing. Early work on cognitive security in remote teams and information ecosystems (Cordes et al. 2020, 2021, 2023) established foundational concepts for information resilience, which this framework extends to artificial agents. Complementary work on Active Inference has demonstrated how cognitive modeling and cognitive science perspectives—including formalization of OODA (Observe-Orient-Decide-Act) loops and multiscale communication dynamics—provide integrative frameworks for understanding agent cognition under adversarial conditions [?]. The OWASP Top 10 for LLM Applications 2025 [?] places prompt injection as the top vulnerability, while the newly released OWASP Top 10 for Agentic Applications [?] specifically addresses autonomous AI systems

Motivation from Recent Deployments I

with “tool misuse, prompt injection, and data leakage’ ’ as primary concerns.

Scale of Deployment (2024–2026):

Motivation from Recent Deployments I

- ▶ Enterprise AI agents processing significant transaction volumes (53% of organizations now deploy RAG and agentic pipelines [?])
- ▶ Autonomous coding assistants with repository write access (GitHub Copilot CVE-2025-53773 demonstrated RCE via prompt injection)
- ▶ Multi-agent orchestrators in infrastructure management contexts

Motivation from Recent Deployments I

Emerging Attack Surface:

Motivation from Recent Deployments I

- ▶ Inter-agent communication channels lack authentication standards, so delegated trust is asserted rather than verified; computational treatments of trust and reputation dynamics model the asymmetry directly—cooperation accrues trust gradually while a violation erodes it sharply [?]
- ▶ Trust delegation mechanisms operate without formal verification—CP-WBFT, a confidence-probe-weighted Byzantine consensus for LLM agents, reports operation at fault rates as high as 85.7% [?]
- ▶ Belief provenance remains largely untracked in production systems—cognitive degradation attacks exploit this gap [?]

Motivation from Recent Deployments I

Limitations of Current Defenses:

Motivation from Recent Deployments I

- ▶ Input/output filtering primarily designed for single-agent architectures—adaptive attacks bypass 12 published defenses with $>90\%$ success [?]
- ▶ Limited standardized frameworks for cognitive integrity verification
- ▶ Byzantine fault tolerance infrequently applied to AI agent systems—emerging work addresses this gap [?]

Motivation from Recent Deployments I

The fundamental constraint is that traditional security models assume a clear boundary between trusted and untrusted components. In multiagent systems, this boundary is fluid—agents must reason about the trustworthiness of other agents' reasoning.

Problem Statement

Problem Statement I

Traditional security models address:

Problem Statement I

- ▶ **Input validation:** Filtering malicious prompts
- ▶ **Output sanitization:** Preventing harmful generations
- ▶ **Access control:** Limiting tool permissions

Problem Statement I

They fail to address:

Problem Statement I

- ▶ **Inter-agent trust:** How should Agent A weight claims from Agent B ?
- ▶ **Belief provenance:** Which beliefs derive from verified vs. adversarial sources?
- ▶ **Coordination integrity:** Can agents be manipulated into malicious consensus?
- ▶ **Temporal persistence:** Do attacks survive context boundaries?
- ▶ **Cognitive integrity:** How can the cognitive systems of today and tomorrow remain flexible and robust amidst change in composition and context?

Research Questions

This paper addresses four fundamental research questions, with emphasis on formal foundations:

RQ1: Taxonomy and Formal Characterization. *What classes of cognitive attacks exist against multiagent systems, and how can they be formally characterized to enable systematic analysis?*

We develop an initial taxonomy spanning epistemic, behavioral, social, and temporal attack dimensions. Crucially, each attack class receives formal definition enabling systematic analysis, composition rules, and detection bounds (sec:attack-taxonomy).

RQ2: Trust Algebra. *How might inter-agent trust be modeled to prevent trust amplification and laundering attacks while enabling legitimate delegation?*

We introduce a trust calculus with bounded delegation (δ^d decay guarantee), prove associativity properties, and establish the no-amplification theorem ensuring that trust cannot be manufactured through delegation chains (sec:trust-calculus).

RQ3: Defense Composition. *How do cognitive defense mechanisms compose, and what guarantees can we provide about layered defense effectiveness?*

We present a defense composition algebra enabling formal reasoning about series and parallel defense arrangements. We prove that orthogonal defenses compose multiplicatively (not additively) for detection rate improvement (sec:defense-composition).

RQ4: Fundamental Bounds. *What are the information-theoretic limits on cognitive attack detection?*

We state the stealth-impact tradeoff theorem, which bounds detection independently of defense implementation, and give an information-geometric argument for its constant. The theorem asserts that attacks cannot simultaneously achieve high impact and complete undetectability; its full proof is deferred, and the proof-status index (S01, sec:proof-status) records which results in this paper are proved and which are asserted (sec:detection-bounds).

Contributions

Contributions I

This paper provides both theoretical foundations and practical mechanisms for cognitive security:

Formal Contributions:

Contributions I

1. **Threat Taxonomy:** A systematic classification of cognitive attacks across epistemic, behavioral, social, and temporal dimensions with formal definitions enabling rigorous analysis (sec:attack-taxonomy)
2. **Trust Calculus:** A mathematical framework for inter-agent trust with bounded delegation (δ^d decay), associativity proofs, and formal guarantees against trust amplification attacks (sec:trust-calculus)
3. **Defense Composition Algebra:** Formal rules for composing security mechanisms with provable detection rate bounds under series and parallel composition (sec:defense-composition)
4. **Information-Theoretic Bounds:** Fundamental limits on attack detection relating stealth constraints to maximum achievable impact, independent of defense implementation (sec:detection-bounds)
5. **Formal Verification:** Model-checked safety properties including belief integrity, trust boundedness, and goal alignment preservation (sec:formal-verification)

Conceptual Contributions:

Contributions I

1. **Cognitive Security Operator Posture:** The proactive defensive stance required when the attack surface spans beliefs, goals, and inter-agent coordination (`sec:operator-posture`)
2. **The Cognitive Integrity Framework (CIF):** An integrated approach combining architectural defenses, runtime monitoring, and Byzantine-tolerant coordination for multiagent systems (`sec:system-model`)

Empirical Validation: Part 2 of this series demonstrates the practical viability of these formal mechanisms across four production architectures, showing that layered cognitive defenses significantly outperform single-mechanism approaches. Practitioner translations, domain-specific applications, and real-world incident analyses appear in the merged Part 3+4 [?].

Paper Organization

The remainder of this paper is structured as follows:

sec:adversary-classes: Threat Model develops a comprehensive adversary taxonomy (Ω_1 – Ω_5) with attack complexity analysis, detectability matrices, and detailed scenarios for each attack class.

sec:system-model: Cognitive Integrity Framework presents the formal foundations of CIF, including system model definitions, cognitive state representations, integrity properties, and the trust calculus.

sec:arch-defenses: Defense Mechanisms describes architectural defenses (cognitive firewalls, belief sandboxing), runtime defenses (tripwires, invariant checking), and coordination defenses (Byzantine consensus, quorum verification).

sec:anomaly-detection: Detection Methods covers anomaly detection algorithms, provenance analysis techniques, and real-time monitoring systems.

sec:formal-verification: Formal Verification proves the main theorems, presents invariant preservation lemmas, and describes model checking configuration.

sec:discussion: Discussion examines limitations, deployment considerations, and connections to related work.

sec:summary: Conclusion summarizes contributions and identifies directions for future research.

Part 2: Experimental Validation A separate, second, companion paper reports empirical results from architecture-aware parametric simulation across 950 attack vectors and four production architectures. Readers seeking detection rates, ablation studies, Bayesian uncertainty quantification, and architecture-specific defense configurations should consult Part 2 (DOI: [10.5281/zenodo.18364128](https://doi.org/10.5281/zenodo.18364128)).

Part 3+4: Practitioner Guidance & Applications The merged third companion paper (*Cognitive Integrity in Practice and Critical Domains*) provides a qualitative review and practitioner's synthesis—accessible-language summaries of the theoretical and empirical findings, operational checklists, risk assessment frameworks, deployment guides, monitoring strategies, cost-benefit analyses—combined with CIF applications across ten high-stakes operational sectors through the integrated CIF-AD-OODA analytical model. Readers interested in domain-specific goal-hijacking defenses, universal attack patterns (FR polarity inversion, constraint relaxation, context boundary violation), or retrospective analysis of documented 2024–2025 AI agent security incidents should consult Part 3 (DOI: 10.5281/zenodo.18364130).

Reading Companion: Where to Find Specific Topics

Reading Companion: Where to Find Specific Topics I

This paper is designed to stand alone as the formal-foundations reference of the series. The table below points readers to the sibling paper and section where each related topic is developed most fully, so that this paper can be used in conjunction with—rather than requiring—its companions.

Reading Companion: Where to Find Specific Topics I

Table 3: Cross-paper navigation from Part 1 topics to sibling developments.

If you want...	...consult...
Empirical detection rates for the defense compositions proved here	Part 2 (DOI: 10.5281/zenodo.18364128), §5 (Results) and S08 (Parametric Analysis)
Ablation studies isolating the contribution of each defense mechanism	Part 2, §5.6 and §5d
Bayesian uncertainty quantification on detection rates	Part 2, §5e
Game-theoretic analysis / Nash equilibrium of CIF vs. adaptive attackers	Part 2, §6 (Discussion)
Model-checking execution (NuSMV, TLA+ traces) for the invariants defined here	Part 2 S04 (Model Checking)
Deployment checklists and operator-posture guidance	Part 3 (DOI: 10.5281/zenodo.18364130), §5 (Deployment Guide), §5b (Incident Response)
Monitoring, drift detection playbooks, cost-benefit analysis	Part 3, §5c–§5d
Case studies showing CIF in complex realistic deployments	Part 3, §6b
Open problems / research directions for practitioners	Part 3 (DOI: 10.5281/zenodo.18364130), §7 (Future Work)
Domain-specific goal-hijacking analyses across ten operational sectors	Part 3, §9c–§9l
Three universal attack patterns (FR Polarity Inversion, Constraint Relaxation, Context Boundary Violation)	Part 3, §10 (Cross-Domain Discussion)
Four novel defense extensions (verification channel separation, active perturbation probing, physics-informed invariants, semiotic decoupling)	Part 3, §9h (Biowarfare), §9j (Trade Wars), §9k (Infrastructure), §9f (Drone Wars)
Retrospective analysis of documented 2024–2025 AI-agent security incidents	Part 3, S03 (Real-World Incidents)

Scope and Limitations

Scope and Limitations I

In scope: Attacks exploiting agent reasoning, trust, and coordination mechanisms in multiagent AI systems.

Out of scope:

Scope and Limitations I

- ▶ Traditional software exploits (buffer overflow, SQL injection, memory corruption)
- ▶ Physical attacks (hardware tampering, side-channel analysis)
- ▶ Supply chain compromise (malicious training data, backdoored models)
- ▶ Cryptographic attacks (we assume secure primitives per `ax:crypto-limit`)

Scope and Limitations I

Assumptions:

Scope and Limitations I

- ▶ Agents communicate over authenticated channels
- ▶ Base model capabilities are not adversarially modified
- ▶ At least one honest orchestrator exists in hierarchical systems